

BRODY ERLANDSON

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EDUCATION

PhD in Statistics, Colorado State University, GPA: 3.87/4.00

Expected: May 2027

M.S. in Data Science, University of Michigan, GPA: 3.97/4.00

April 2022

B.S. in Mathematics, Eastern Michigan University, GPA: 3.95/4.00

December 2019

Minor in Philosophy, **Honors**: Dean's list; Summa Cum Laude

EXPERIENCE

Graduate Teaching and Research Assistant

Aug 2022 – Present

Colorado State University

Fort Collins, CO

- Supported instruction in courses such as Machine Learning, Bayesian Statistics, and Computational and Simulation Methods.

Algorithms (Machine Learning) Engineer Intern

May 2023 – Aug 2023 & May 2024 – Dec 2024

KLA

Ann Arbor, MI

- Researched and adapted deep learning classification methods for noisy label datasets using KLA data.
- Developed uncertainty quantification and model evaluation workflows to assess model performance on new data and implemented resulting methods in the codebase.

Student Research Assistant II

March 2021 – Aug 2022

University of Michigan

Ann Arbor, MI

- Cleaned and prepared Nielsen Consumer Panel data and conducted unsupervised analyses.

Lecturer (Part-Time)

May 2022 – Aug 2022

Washtenaw Community College

Ann Arbor, MI

- Taught two sections of Linear Algebra, including one in-person and one online section.

Graduate Student Instructor

Aug 2021 – April 2022

University of Michigan

Ann Arbor, MI

- Supported instruction for Introduction to Statistics and Linear Regression Analysis courses.

SELECTED RESEARCH & PROJECTS

Research In Progress

- Developing methods to estimate latent heterogeneity of longitudinal causal effects for micro-randomized trials.

A Bayesian Functional Concurrent Zero-Inflated Dirichlet-Multinomial Regression Model with Application to Infant Microbiome First Author Paper

- Developed a regression model, FunC-ZIDM, to model time-varying relations between observed covariates and microbial taxa while accounting for zero-inflation, compositionality, and repeated measures.

Assessing Uncertainty and Methods of Evaluation in Deep Learning Classification Models KLA internship project

- Researched and modified state-of-the-art uncertainty quantification and evaluation metrics for deep learning classification techniques for KLA data. The methods were used to assess the performance of models on new data.

Deep Learning Classification with Noisy Labels KLA internship project

- Researched and adapted methods utilizing semi- and self-supervised learning alongside advanced CNNs. Achieved accuracies within 3–5% of those obtained on clean datasets despite up to 20% label noise.

SKILLS

Statistical Methods

Bayesian modeling, causal inference, probabilistic machine learning, multivariate outcomes, supervised and unsupervised learning, statistical computing.

Programming

C++, R, Python, SQL, SAS.

Computing & Tools

High-performance computing, Linux command line, data manipulation, Git/GitHub, L^AT_EX.